

Question Paper for 1st Periodical Examination (March 2026) (Session Jan-June 2026)

Time: One and a half hours

Max Marks:30

Note: Attempt all questions.

Date: 12.03.2026 (10-11:30AM)

CO	COURSE OUTCOMES
CO1	Design and implement machine learning solutions to classification, regression and clustering problems.
CO2	Evaluate and interpret the results of the different ML techniques.
CO3	Design and implement various machine learning algorithms in a range of Real-world applications.

Q No.	Questions	Marks	Target CO	Target KL
1	<i>Attempt all:</i>	2*5=10	1,2,3	
1.a	What is the limitation of a single-layer perceptron model?	1	CO1	L2
1.b	Name the loss function used for the linear regression task.	1	CO1	L2
1.c	Compare a biological neuron and an artificial neuron.	3	CO1	L3
2.	Name the first computational neural network. Explain its characteristics. How it is helpful for implementing basic logic gates?	1+2+2	CO1	L3
3	Explain back propagation with the help of an example.	4	CO2	L3
4.	Explain any 4 types of artificial neural networks.	1.5x4=6	CO2	L2
5.	Write the expression for cost function for a dataset for the following, with a description of each variable. And an example for each.:	2x5=10	CO1	L3
	a. Root Mean Squared Error			
	b. Huber Loss			
	c. Log Loss			
	d. Categorical Cross-Entropy			
	e. Hinge Loss			

KL: Knowledge Level: L1 - Remember L2 - Understand L3 - Apply L4 - Analyze L5 - Evaluate L6 - Create
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